Inorganic Cumulative Exam
January 16 1998

Answer 4 out of the following 5 questions.

1. Zeolites are of significant commercial importance.
   a. What is the general chemical composition of these important materials?
   b. Describe one of the features of zeolites which is important to their industrial applications.
   c. Give an example of a commercial application.

2. In the recent seminar by Scott Oliver work with aluminophosphates was discussed. Dr. Oliver described two approaches to the synthesis of structured aluminophosphates which had not been previously explored. Describe and explain the significance of each.

3. Simple phosphates such as sodium dihydrogen phosphate form a large number of polyphosphates upon heating. Give some examples of expected products. Which of these is commercially important?

4. Researchers in the materials science area are often interested in creating molecular pieces which can be used to assemble larger structures. "Zintl phases" or "Zintl ions" might be considered molecular building blocks? Give an example of a Zintl ion.

5. Silicates offer a very large variety of three dimensional structures as exemplified by naturally occurring minerals. Show how the "SiO_4 subunits" can be organized into:
   a. rings
   b. linear chains
   c. sheets